Anamika Koonakkampilly Sunilal

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LinkedIn | GitHub

MSc Artificial Intelligence graduate with project experience in predictive modelling, multimodal deep learning, and Generative AI. Built a fashion classifier achieving **91.4% held-out test accuracy across 90 categories** and developed explainable churn models, RAG applications, and interactive dashboards.

Technical Skills

Programming Languages: Python, SQL, Java, C++, JavaScript
AI and Machine Learning: Deep Learning, NLP, Computer Vision, RAG, Reinforcement Learning, XAI
Frameworks and Libraries: TensorFlow, PyTorch, Scikit-learn, XGBoost, SHAP, Hugging Face, OpenCV
Databases and Analytics: PostgreSQL, Excel, Pandas, NumPy, Matplotlib, Seaborn
Systems and Development: Git, GitHub, Linux, Streamlit, Django

Education

Dublin Business School, Ireland Sep 2025 – Sep 2026
MSc Artificial Intelligence
Dissertation: Explainable AI for telecom customer churn prediction.

St. Joseph's College (Autonomous), India Aug 2022 – Mar 2025
Bachelor of Computer Applications

Selected Projects

Multimodal Fashion Product Classification GitHub
TensorFlow, Keras, EfficientNetV2, DistilBERT

- Designed a late-fusion classifier combining EfficientNetV2 image and DistilBERT text encoders for **44,020 fashion products across 90 categories**.
- Implemented preprocessing, tokenisation, training, and inference in TensorFlow, achieving **91.4% held-out test accuracy**.

Explainable Telecom Customer Churn Prediction
Python, Scikit-learn, XGBoost, TensorFlow, SHAP

- Developed a churn prediction pipeline on **7,043 customer records** using preprocessing, feature engineering, SMOTE, and an 80:20 stratified split; benchmarked eight configurations across Logistic Regression, Random Forest, XGBoost, and MLP.
- Tuned models using GridSearchCV and five-fold cross-validation, achieving **XGBoost CV F1 of 0.7147**; applied SHAP for global and customer-level explanations.

LLM RAG Knowledge Assistant GitHub
Python, Hugging Face, NLP, Vector Retrieval

- Developed a research-document RAG pipeline integrating document processing, semantic retrieval, LLM generation, query generation, and citation verification to support source-grounded answers.

Adaptive Traffic Signal Control GitHub | Live Demo
Python, Streamlit, Q-Learning, Monte Carlo

- Implemented Q-Learning and Monte Carlo agents for adaptive traffic signal control; developed a Streamlit dashboard comparing reward, throughput, queue length, waiting time, and signal switching.

Experience and Leadership

Machine Learning and AI Intern May 2024 – Jun 2024
ICT Academy of Kerala

- Applied KNN, linear regression, and K-means using Python and Scikit-learn to analyse marketing campaign data, identify customer segments, and examine behavioural patterns.

Technical Lead Jan 2024 – Mar 2025
IEDC, St. Joseph's College (Autonomous)

- Coordinated **10+ technical events for over 300 participants** and mentored **15+ student teams**.
- Managed IEDC digital platforms and won **Ideathon 3.0 against 25 teams** with an Agritech crop yield forecasting solution.